

A solution of the system is a list $(s_1, s_2, \dots, s_n)$ of numbers that makes each equation a true statement when the values $s_1, \dots, s_n$ are substituted for $x_1, \dots, x_n$, respectively. For instance, $(5, 6, 5, 3)$ is a solution of system (2) because, when these values are substituted in (2) for x_1, x_2, x_3 , respectively, the equations simplify to $8=8$ and $-7=-7$.

The set of all possible solutions is called the solution set of the linear system. Two linear systems are called equivalent if they have the same solution set. That is, each solution of the first system is a solution of the second system, and each solution of the second system is a solution of the first.

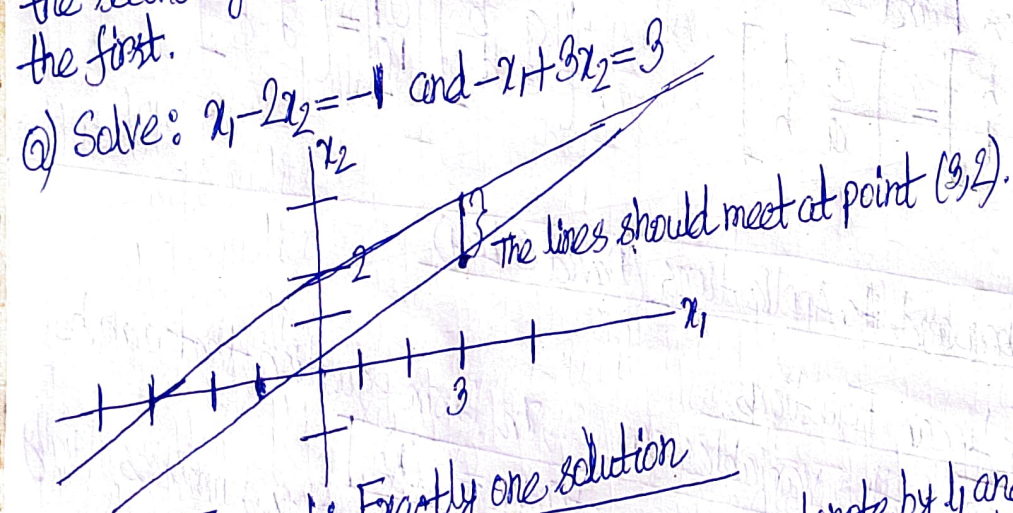


Figure 1: Exactly one solution

The graphs of these equations are lines, which we denote by l_1 and l_2 . A pair of numbers (x_1, x_2) satisfies both equations in the system if and only if the point (x_1, x_2) lies on both l_1 and l_2 . In the system above, the solution is the single point $(3, 2)$, as you can easily verify. See Figure 1.

Of course, two lines need not intersect in a single point — they could be parallel, or they could coincide and hence intersect at every point on the line. Figure 2 shows the graphs that correspond to the following systems:

